

Shadow Prices Explained (What Are Lambda Values?)

ECO 310 - Intuitive Explanation

February 23, 2026

1 What Is a Shadow Price?

Simple Definition

A **shadow price** is how much your objective function would improve if you could relax a constraint by a tiny bit.

In this problem: **Shadow price = How much would your cost decrease if you could slightly relax a constraint?**

The λ values in your Lagrangian ARE the shadow prices!

- λ_1 = shadow price of constraint $x_1 \geq 0$
- λ_2 = shadow price of constraint $x_2 \geq 0$
- λ_3 = shadow price of constraint $x_1 + x_2 \geq u^*$

2 Concrete Example: Coffee and Tea

Let's use actual numbers to understand shadow prices.

Setup:

- Coffee costs \$3 per cup ($p_1 = 3$)
- Tea costs \$5 per cup ($p_2 = 5$)
- You need utility 10 ($u^* = 10$)

Optimal solution: Buy 10 coffees, 0 teas (cost = \$30)

So: $x_1 = 10$, $x_2 = 0$

2.1 Shadow Price of λ_1 (Coffee Nonnegativity)

Constraint: $x_1 \geq 0$ (can't buy negative coffee)

Question: How much would your cost decrease if you could buy $x_1 = -0.01$ (slightly negative)?

Answer: Your cost wouldn't decrease at all! You're already buying 10 coffees, so being allowed to buy negative amounts doesn't help you.

Therefore: $\lambda_1 = 0$ (this constraint is not binding, relaxing it doesn't help)

Interpretation

$\lambda_1 = 0$ means: The constraint $x_1 \geq 0$ is NOT binding. You're not "stuck" at $x_1 = 0$, so relaxing this constraint has zero value to you.

2.2 Shadow Price of λ_2 (Tea Nonnegativity)

Constraint: $x_2 \geq 0$ (can't buy negative tea)

Question: How much would your cost decrease if you could buy $x_2 = -0.01$ (slightly negative)?

Answer: This would help! If you could "buy negative tea" (i.e., sell tea you don't have and get money back), you'd want to do it.

Let's calculate:

- If you could set $x_2 = -1$ (sell 1 unit of tea): you'd get +\$5
- But you'd need to replace that utility: buy 1 more coffee for \$3
- Net savings: \$5 - \$3 = \$2

Therefore: $\lambda_2 = 2$ (relaxing this constraint by 1 unit would save you \$2)

Interpretation

$\lambda_2 = 2$ means: The constraint $x_2 \geq 0$ IS binding. You're stuck at $x_2 = 0$, and if you could relax this constraint (buy negative tea), each unit would save you \$2.

Why \$2? Because tea costs \$5 but coffee (the alternative) costs \$3, so the price difference is \$2.

2.3 Shadow Price of λ_3 (Utility Constraint)

Constraint: $x_1 + x_2 \geq u^* = 10$ (must reach utility 10)

Question: How much would your cost decrease if you only needed utility 9.99 instead of 10?

Answer: You'd need 0.01 fewer coffees, saving $0.01 \times \$3 = \0.03

Per unit of utility: Cost would decrease by \$3

Therefore: $\lambda_3 = 3$ (marginal cost of utility is \$3)

Interpretation

$\lambda_3 = 3$ means: Each additional unit of utility costs you \$3 more.

Why \$3? Because you're buying the cheaper good (coffee at \$3), so increasing utility by 1 unit means buying 1 more coffee for \$3.

This is also called the marginal cost of utility.

3 General Pattern for Shadow Prices

How Shadow Prices Work

If constraint is NOT binding ($\lambda = 0$):

- You're not stuck at the constraint boundary
- Relaxing the constraint doesn't help you
- Example: $x_1 = 10 > 0$, so $\lambda_1 = 0$

If constraint IS binding ($\lambda > 0$):

- You're stuck at the constraint boundary
- Relaxing the constraint would help you
- The λ value tells you exactly how much it would help
- Example: $x_2 = 0$, so $\lambda_2 > 0$

4 Complementary Slackness

Shadow prices are connected to complementary slackness:

Complementary Slackness Rules

For each constraint and its shadow price λ :

Either:

1. The constraint is binding (exactly satisfied) and $\lambda \geq 0$
2. OR the constraint is NOT binding (slack) and $\lambda = 0$

They can't both be positive at the same time!

Mathematically: (constraint slack) $\times$ (shadow price) = 0

4.1 Examples

Constraint $x_1 \geq 0$:

- If $x_1 = 0$ (binding): λ_1 can be > 0
- If $x_1 > 0$ (not binding): $\lambda_1 = 0$
- Complementary slackness: $x_1 \cdot \lambda_1 = 0$

Constraint $x_2 \geq 0$:

- If $x_2 = 0$ (binding): λ_2 can be > 0
- If $x_2 > 0$ (not binding): $\lambda_2 = 0$

- Complementary slackness: $x_2 \cdot \lambda_2 = 0$

Constraint $x_1 + x_2 \geq u^*$:

- If $x_1 + x_2 = u^*$ (binding): λ_3 can be > 0
- If $x_1 + x_2 > u^*$ (not binding): $\lambda_3 = 0$
- Complementary slackness: $(x_1 + x_2 - u^*) \cdot \lambda_3 = 0$

5 How Shadow Prices Are Used in Problem 2b

In Problem 2b, you're given a guess:

- $x_1 = 0$ (binding) $\Rightarrow \lambda_1$ can be positive
- $x_2 > 0$ (not binding) $\Rightarrow \lambda_2 = 0$
- $x_1 + x_2 = u^*$ (binding) $\Rightarrow \lambda_3$ can be positive

5.1 Using Complementary Slackness to Simplify

Step 1: Since $x_2 > 0$, we know $\lambda_2 = 0$

Step 2: The FOC for x_2 is:

$$-p_2 + \lambda_2 + \lambda_3 = 0 \tag{1}$$

Substitute $\lambda_2 = 0$:

$$-p_2 + \lambda_3 = 0 \quad \Rightarrow \quad \lambda_3 = p_2 \tag{2}$$

Step 3: The FOC for x_1 is:

$$-p_1 + \lambda_1 + \lambda_3 \leq 0 \tag{3}$$

Substitute $\lambda_3 = p_2$:

$$-p_1 + \lambda_1 + p_2 \leq 0 \quad \Rightarrow \quad \lambda_1 \leq p_1 - p_2 \tag{4}$$

At optimum, this typically binds:

$$\lambda_1 = p_1 - p_2 \tag{5}$$

Step 4: Check validity. For $\lambda_1 \geq 0$:

$$p_1 - p_2 \geq 0 \quad \Rightarrow \quad p_1 \geq p_2 \tag{6}$$

So this guess is valid when good 1 is at least as expensive as good 2.

6 Economic Interpretation of the Solution

From our example where $p_1 = 3$, $p_2 = 5$:

- $\lambda_1 = 0$: Nonnegativity on coffee is not constraining. You're happy buying coffee.
- $\lambda_2 = p_2 - p_1 = 5 - 3 = 2$: Nonnegativity on tea IS constraining. If you could "buy negative tea" (sell tea back), each unit would save you \$2 (the price difference).
- $\lambda_3 = p_1 = 3$ (or p_2 depending on which good you buy): The marginal cost of utility. Increasing u^* by 1 unit costs \$3 more (the price of the cheaper good).

7 Summary

Shadow Prices in Plain English

What they are: The λ values measure "how much would I save if I could relax this constraint?"

How to use them:

1. If you're told a constraint is binding, the shadow price can be positive
2. If you're told a constraint is NOT binding, the shadow price must be zero
3. Use complementary slackness to set certain λ values to zero
4. This simplifies the optimality conditions into solvable equations

Economic meaning:

- $\lambda_i = 0$: Constraint i is not limiting you
- $\lambda_i > 0$: Constraint i is limiting you, and λ_i tells you the value of relaxing it